

Alexander Steinmaurer - Publications

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Peer-reviewed publications

2026

Towards Improving CS Students' Generative AI Literacy

Bruno Pereira Cipriano, Olga Petrovska, Nuno Pombo, Lina Battestilli, Laura Farinetti, Richard Glassey, Maria Kasinidou, Olakunle Olayinka, Anshul Shah, [Alexander Steinmaurer](#), Ramalakshmi Vaidhiyanathan, & Weichert James

Proceedings of the 31st ACM Conference on Innovation and Technology in Computer Science Education V. 2, pp. 789–790

Conference, 2026

DOI: [10.1145/3803401.3812055](https://doi.org/10.1145/3803401.3812055)

2025

GenAI Integration in Upper-Level Computing Courses

Dennis J. Bouvier, Bruno Pereira Cipriano, Richard Glassey, Raymond Pettit, Emma Anderson, Anastasiia Birillo, Ryan E. Dougherty, Orit Hazzan, Olga Petrovska, Nuno Pombo, Ebrahim Rahimi, Charanya Ramakrishnan, [Alexander Steinmaurer](#), Shubbhi Taneja, Muhammad Usman, Annapurna Vadaparty, & Govindha Ramaiah Yeluripati

Proceedings of the 30th ACM Conference on Innovation and Technology in Computer Science Education V. 2, pp. 691–692

Conference, 2025

DOI: [10.1145/3724389.3731276](https://doi.org/10.1145/3724389.3731276)

Real-World Study of the Security of Educational Test Systems

Stefan Gast, Sebastian Daniel Felix, [Alexander Steinmaurer](#), Jonas Juffinger, & Daniel Gruss

IEEE European Symposium on Security and Privacy Workshops (EuroS&P Workshops), pp. 588–599

Conference, 2025

DOI: [10.1109/EUROSPW67616.2025.00074](https://doi.org/10.1109/EUROSPW67616.2025.00074)

The Rest of the Robots: Generative AI in Post-introductory Computing Education

Dennis J. Bouvier, Bruno Pereira Cipriano, Richard Glassey, Olga Petrovska, Emma Anderson, Anastasiia Birillo, Ryan E. Dougherty, Raymond Pettit, Nuno Pombo, Ebrahim Rahimi, Charanya Ramakrishnan, [Alexander Steinmaurer](#), Shubbhi Taneja, Muhammad Usman, & Annapurna Vadaparty

Proceedings of the 2025 Working Group Reports on Innovation and Technology in Computer Science Education, pp. 61–107

Conference, 2025

DOI: [10.1145/3760545.3783970](https://doi.org/10.1145/3760545.3783970)

To Complete or Not to Complete: Prediction and Classification of Dropouts in a CS1 Course

Florian Schnedlitz, David Kerschbaumer, [Alexander Steinmaurer](#), David Lowe, & Christian Gütl

IEEE Global Engineering Education Conference (EDUCON), pp. 1–7

Conference, 2025

DOI: [10.1109/EDUCON62633.2025.11016432](https://doi.org/10.1109/EDUCON62633.2025.11016432)

2024

Conversational Agents, Virtual Worlds, and Beyond: A Review of Large Language Models Enabling Immersive Learning

Andreas Pester, Ahmed Tammaa, Christian Gütl, [Alexander Steinmaurer](#), & Samir Abou El-Seoud
IEEE Global Engineering Education Conference (EDUCON), pp. 1–6
Conference, 2024

DOI: [10.1109/EDUCON60312.2024.10578895](https://doi.org/10.1109/EDUCON60312.2024.10578895)

Immersive Learning in History Education: Exploring the Capabilities of Virtual Avatars and Large Language Models

[Alexander Steinmaurer](#), Andreas Dengel, Mario Comanici, Josef Buchner, Josef Memminger, & Christian Gütl
Immersive Learning Research Network - 10th International Conference on Immersive Learning, 2271, pp. 363–374
Conference, 2024

DOI: [10.1007/978-3-031-80475-5_26](https://doi.org/10.1007/978-3-031-80475-5_26)

Iterative Student Program Planning using Transformer-Driven Feedback

Elijah Rivera, [Alexander Steinmaurer](#), Kathi Fisler, & Shriram Krishnamurthi
Proceedings of the 2024 ACM Conference on Innovation and Technology in Computer Science Education V. 1
Conference, 2024

DOI: [10.1145/3649217.3653607](https://doi.org/10.1145/3649217.3653607)

2023

Analyzing Social Networks and Learning Content on a Discussion Forum of an Introductory Programming MOOC in Higher Education

[Alexander Steinmaurer](#), Ratko Savic, Johannes Krugel, & Christian Gütl
IEEE Learning with MOOCS, pp. 1–6
Conference, 2023

DOI: [10.1109/LWMOOCS58322.2023.10305937](https://doi.org/10.1109/LWMOOCS58322.2023.10305937)

CPR Emergency Assistance Through Mixed Reality Communication

Manuel Rebol, [Alexander Steinmaurer](#), Florian Gamillscheg, Krzysztof Pietroszek, Christian Gütl, Claudia Ranniger, Colton Hood, Adam Rutenberg, & Neal Sikka
Augmented Intelligence and Intelligent Tutoring Systems, 13891, pp. 415–429
Conference, 2023

DOI: [10.1007/978-3-031-32883-1_38](https://doi.org/10.1007/978-3-031-32883-1_38)

Research Agenda 2030: The Great Questions of Immersive Learning Research

Andreas Dengel, [Alexander Steinmaurer](#), Lea Marie Müller, Melanie Platz, Minjuan Wang, Christian Gütl, Andreas Pester, & Leonel Morgado
Immersive Learning Research Network - 9th International Conference, 1904, pp. 161–172
Conference, 2023

DOI: [10.1007/978-3-031-47328-9_12](https://doi.org/10.1007/978-3-031-47328-9_12)

2022

Analyzing Behavioral Patterns in an Introductory Programming MOOC at University Level

[Alexander Steinmaurer](#), Christoph Schatz, Johannes Krugel, & Christian Gütl
IEEE Learning with MOOCS, pp. 114–119
Conference, 2022

DOI: [10.1109/LWMOOCS53067.2022.9927886](https://doi.org/10.1109/LWMOOCS53067.2022.9927886)

Developing and Evaluating a Multiplayer Game Mode in a Programming Learning Environment

[Alexander Steinmaurer](#), David Eckhard, Julius Dreveny, & Christian Gütl
8th International Conference of the Immersive Learning Research Network, pp. 1–8
Conference, 2022

DOI: [10.23919/ILRN55037.2022.9815973](https://doi.org/10.23919/ILRN55037.2022.9815973)

Implementation and Experiences of a Flipped Lecture Hall - A Fully Online Introductory Programming Course

Alexander Steinmaurer & Christian Gütl

Learning in the Age of Digital and Green Transition, 633, pp. 832–843

Conference, 2022

DOI: [10.1007/978-3-031-26876-2_79](https://doi.org/10.1007/978-3-031-26876-2_79)

Learning Security Awareness in Email Communication Using a Platform for Digital Skill Teaching

Alexander Steinmaurer, Azra Bajramovic, Daniel Pollhammer, & Christian Gütl

IEEE International Conference on Teaching, Assessment and Learning for Engineering (TALE), pp. 38–45

Conference, 2022

DOI: [10.1109/TALE54877.2022.00015](https://doi.org/10.1109/TALE54877.2022.00015)

Work-in-Progress-Volumetric Communication for Remote Assistance Giving Cardiopulmonary Resuscitation

Manuel Rebol, Krzysztof Pietroszek, Claudia Ranniger, Colton Hood, Adam Rutenberg, Neal Sikka, Alexander Steinmaurer, & Christian Gütl

8th International Conference of the Immersive Learning Research Network, pp. 1–3

Conference, 2022

DOI: [10.23919/ILRN55037.2022.9815989](https://doi.org/10.23919/ILRN55037.2022.9815989)

2021

Beyond Gaming: The Potential of Twitch for Online Learning and Teaching

Johanna Pirker, Alexander Steinmaurer, & Aleksandar Karakas

Proceedings of the 26th ACM Conference on Innovation and Technology in Computer Science Education V. 1, pp. 74–80

Conference, 2021

DOI: [10.1145/3430665.3456324](https://doi.org/10.1145/3430665.3456324)

Designing and Developing a Learning Analytics Platform for the Coding Learning Game sCool

Alexander Steinmaurer, Anil Kumar Tilanthe, & Christian Gütl

New Realities, Mobile Systems and Applications, 411, pp. 547–558

Conference, 2021

DOI: [10.1007/978-3-030-96296-8_49](https://doi.org/10.1007/978-3-030-96296-8_49)

Engagement in In-Game Questionnaires - Perspectives from Users and Experts

Alexander Steinmaurer, Martin Sackl, & Christian Gütl

7th International Conference of the Immersive Learning Research Network, pp. 1–7

Conference, 2021

DOI: [10.23919/ILRN52045.2021.9459373](https://doi.org/10.23919/ILRN52045.2021.9459373)

2020

Immersively Learning Object Oriented Programming Concepts With sCool

Chanelle Kaith Mosquera, Alexander Steinmaurer, Christian Eckhardt, & Christian Guetl

6th International Conference of the Immersive Learning Research Network, pp. 124–131

Conference, 2020

DOI: [10.23919/ILRN47897.2020.9155144](https://doi.org/10.23919/ILRN47897.2020.9155144)

sCool: Impact on Human-Computer Interface Improvements on Learner Experience in a Game-Based Learning Platform

Martin Sackl, Alexander Steinmaurer, Christopher Cheong, France Cheong, Justin Filippou, & Christian Gütl

Educating Engineers for Future Industrial Revolutions, 1328, pp. 439–451

Conference, 2020

DOI: [10.1007/978-3-030-68198-2_41](https://doi.org/10.1007/978-3-030-68198-2_41)

2019

sCool - Game-Based Learning in Computer Science Class: A Case Study in Secondary Education

Alexander Steinmaurer, Johanna Pirker, & Christian Gütl

International Journal of Engineering Pedagogy, 9(2), pp. 35–50

Journal, 2019

DOI: [10.3991/ijep.v9i2.9942](https://doi.org/10.3991/ijep.v9i2.9942)

2018

sCool - Game Based Learning in STEM Education: A Case Study in Secondary Education

Alexander Steinmaurer, Johanna Pirker, & Christian Gütl

The Challenges of the Digital Transformation in Education, 916, pp. 614–625

Conference, 2018

DOI: [10.1007/978-3-030-11932-4_58](https://doi.org/10.1007/978-3-030-11932-4_58)

Preprints

2025

Automated Feedback on Student-Generated UML and ER Diagrams Using Large Language Models

Sebastian Gürtl, Gloria Schimetta, David Kerschbaumer, Michael Liut, & Alexander Steinmaurer

CoRR, abs/2507.23470

Other, 2025

DOI: [10.48550/arXiv.2507.23470](https://doi.org/10.48550/arXiv.2507.23470)

DreamLLM-3D: Affective Dream Reliving using Large Language Model and 3D Generative AI

Pinyao Liu, Keon Ju M. Lee, Alexander Steinmaurer, Claudia Picard-Deland, Michelle Carr, & Alexandra Kitson

CoRR, abs/2503.16439

Other, 2025

DOI: [10.48550/arXiv.2503.16439](https://doi.org/10.48550/arXiv.2503.16439)